CSS 423 - Internet of Things: Experiment-1

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Experiment-1: Automatic Light Controlling System

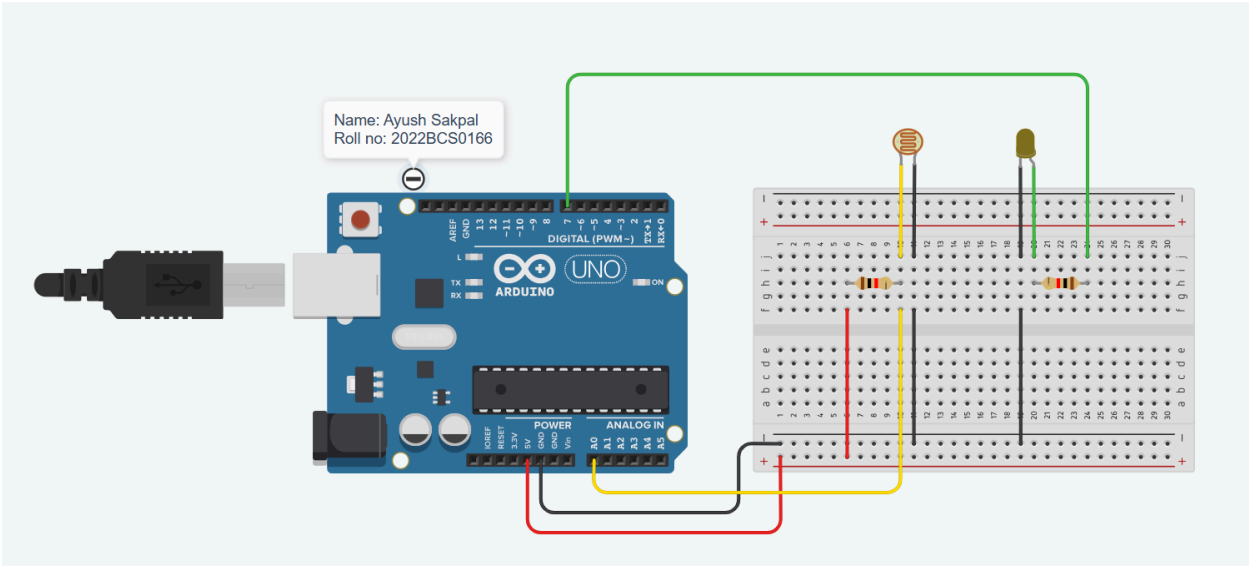
Objective

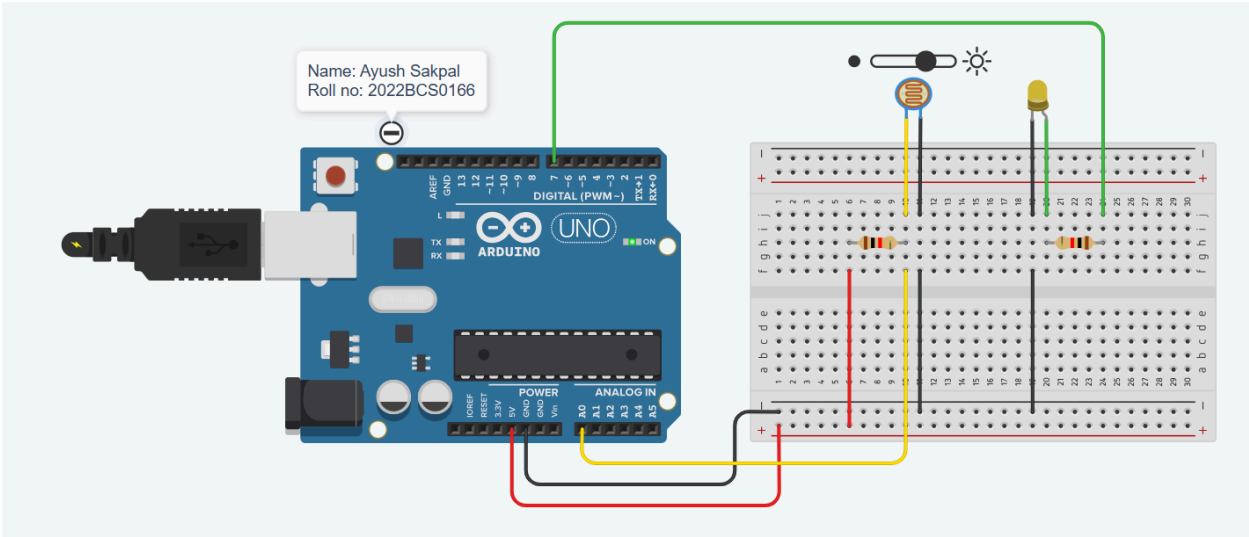
Create an automatic light controlling system using LDR sensor and LEDs.

Components Required

1. Arudino UNO
2. LDR (Photoresistor)
3. 1 x LED
4. 2 x Resistors (1kΩ)
5. Mini Breadboard
6. Connecting Wires

Circuit Diagram





Circuit Explanation

In this circuit, an LDR (Light Dependent Resistor) is used to detect light. The resistance of the LDR changes based on the amount of light falling on it.

The LDR is connected to the analog pin A0 of the Arduino using a resistor, forming a voltage divider circuit. This allows the Arduino to read different values depending on light intensity.

- When there is more light, the LDR resistance becomes low.
- When there is less light or darkness, the LDR resistance becomes high.

The Arduino reads the light value using analogRead(A0).

An **LED** is connected to **digital pin 7** through a resistor.

This LED acts as the output light.

The Arduino compares the LDR value:

- If the value is less than 500 (dark condition), the LED is turned ON.
- If the value is greater than 500 (bright condition), the LED is turned OFF.

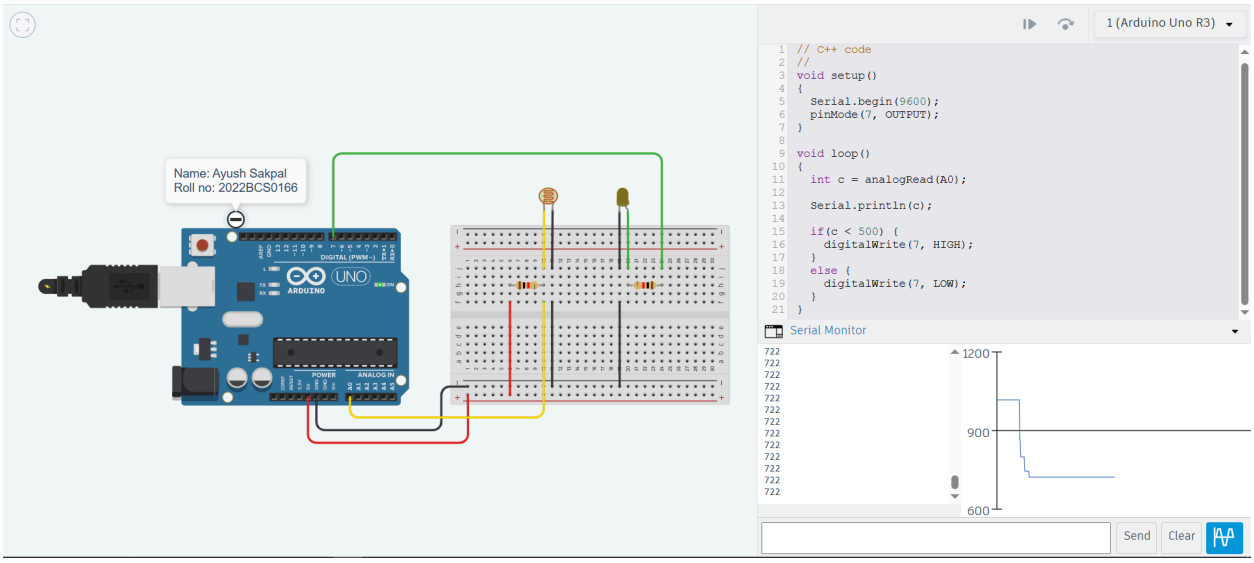
Thus, the LED automatically turns ON in darkness and turns OFF in light.

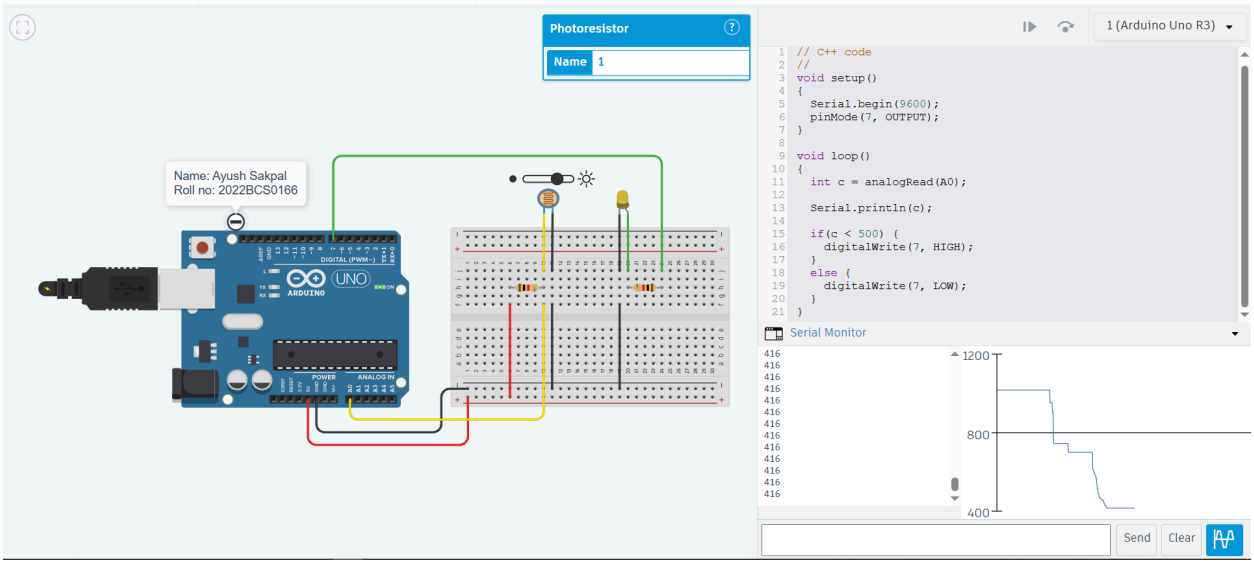
Code

```
void setup()
{
  Serial.begin(9600);
  pinMode(7, OUTPUT);
}

void loop()
{
  int c = analogRead(A0);

  Serial.println(c);

  if(c < 500) {
  digitalWrite(7, HIGH);
  }
  else {
  digitalWrite(7, LOW);
  }
}
```

Result

The automatic light controlling system worked successfully.
When the surrounding area was dark, the LED turned ON automatically.
When sufficient light was present, the LED turned OFF automatically.
The light intensity values were also displayed on the Serial Monitor.





Conclusion

This experiment shows how an Arduino can be used to control a light automatically using an LDR sensor. It helped in understanding how sensors work,

how analog values are read using Arduino, and how an output device like an LED can be controlled based on sensor input.